

The Intensity of the Cultural Layer

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The development of the modern archaeology, as it is conventionally deemed, put into agenda elaboration of issues concerning exact archaeological notions and terms. Now it is clearly evident that in the course of the process of practical and theoretical researches the currently existing essential definitions of some notions and terms are pointless. At present they are established in archaeological ‘language’, factually representing linguistic basics conceptualized at a low level of a direct communication.

The presented work deals with a limited issue of the layer theory – definition of a quantitative measurement of spreading of the archaeological material in a layer. Correspondingly, its objective is an exact and specific definition of the previously mentioned measurement in the cultural layers of the settlements.

No distinctly defined criterion exists in archaeology for identification of an amount of materials spreading in a cultural layer. Correspondingly, no unified methodologically justified base for estimation of the materials spreading in cultural layers of various sites exists. The necessity of introduction of a similar quantitative measure is always required. This requirement reveals itself in the phrases: ‘the layer contains few materials’ or ‘the layer abounds with materials’, ‘the material is scarce’, etc. However, for the reason that these phrases have no quantitative measurement estimated from a common starting point, they in fact represent meaningless and practically useless notions for submitting information, expression of which they should convey. More precise explanation of this idea reads as follows: Necessity of juxtaposition of certain archaeological sites and layers frequently arises at various stages of the archaeological and historical researches. Each fact – life expectancy of the population, intensity of its cultural or economic life, etc. – acquires a distinct definition only in case of juxtaposition carried out against a methodologically correct background, i.e. in case of juxtaposition of facts (e.g. for definition of life intensity at the multilayered settlement, it is necessary to identify content of materials in layers dated from various periods of time and to compare them with others).

However, in the course of such juxtaposition/comparison certain kind of resistance appears, since, as a rule, settlements or habitation layers, regardless of being synchronous or asynchronous, always differ from each other by their quantitative parameters – areas occupied by them are different as well as thickness of various layers and amounts of finds. On the other hand, in each single settlement or layer all these parameters are inseparably integrated. Therefore consideration of a single artefact in isolation from other ones means one-sided discussion of the settlement or the layer. Consequently, comparison of settlements (or layers dated from a certain period of time) on the basis of existing parameters (for instance, on the grounds of amounts of materials) is absolutely unjustified from the methodological point of view, since in separate cases these amounts could be depended upon an area occupied by a settlement (or a layer) or upon

thickness of a layer. Thus, naturally, search for such a criterion, which could manage to arrange and envisage the links, existing between all these parameters, becomes necessary and at the same time would give us such an amount of archaeological material, existing in the layer, which would express life intensity of the society at a concrete site, since amount of material evidenced in the layer closely depends upon the intensity of human activities in manufacture, household, trade, economics or culture.

Correspondingly, analyzing materials spreading in a layer by means of a unified criterion can turn into an objective basis, into an actually comparable unit for the definition of real intensity of human activity at each single site of similar character.

Elaboration of the aforementioned criteria requires factually indefinite time span for observations in the course of archaeological excavations performed in practice. Besides it means investigation of a layer maximally rich in materials, which on its own way creates a basis for possible imprecision to such an extent that the amount of maximal intensity proper needs definition. Correspondingly, elaboration of a criterion is possible only by means of an experimental way. But, a problem arises here – what should be considered as a starting point – maximal or minimal amount of materials in a layer. Besides, what quantitative amounts could correspond to the former or the latter ones.

It is absolutely obvious that existence of a minimal amount of the material in the layer will be inappropriate as the minimal amount expresses a certain value – a zero, meaning that no material is in the layer and that such a layer does not deserve to be subjected to special archaeological excavations. It no more represents a cultural layer. However, notwithstanding this, if we assume that existence of material in the cultural layer corresponds to 0%, then it could be assumed that one or ten or any number corresponds to 1, 10 or any amount of percentage, i.e. this criterion actually turns to be infinite, and, correspondingly, absolutely useless.

Thus, it is obvious that a maximal amount of materials evidenced in the given layer (i.e. this amount has to correspond to a maximal, 100%-amount) should be considered as a starting point for elaboration of the criterion. It turns this criterion proper into a finite, limited unit, at the same time turning any amount of the material definable within the frames of 0 and 100%.

Each layer, from the quantitative point of view, is characterized with three main components: horizontal and vertical spreading (i.e. its area and thickness) and amount of the material (i.e. intensity of a layer). For the reason that all these components and their presentation at various sites are variable, a cubic meter of layer has been accepted as a standard for definition of the maximal amount of the material in the layer. First of all, usage of such dimensional volume is optimal as horizontal and vertical spreading of a layer mostly exceeds the standard and, secondly, this dimension is much more convenient for carrying out all kinds of mathematical operations and, correspondingly, it is easy for practical usage.

In the course of studying the excavated archaeological sites from this point of view and in the process of experiments, it became evident that amount of the material in a layer drastically differs in cases of the sites dated from the Stone Age and the subsequent periods of time. This difference has been clearly confirmed in the course of the experiment. It has been evidenced that, for instance, a number of Upper Paleolithic flakes in a volume of a cubic meter layer theoretically can reach 172 000 samples (the experiment was performed on the basis of the materials obtained at Dzudzuana. Besides, it is noteworthy that an actually maximal amount of the material was evidenced at the same site, so far. Namely, 1,746

flakes were discovered in a layer with its thickness corresponding to 10 cm and its area covering 1 meter. These data calculated in case of one cubic meter, corresponded to 17 460 artefacts. In addition, it should be noted that at the Mesolithic site of Edzanis grotto, which is also distinguishable for the abundance of its material, in accordance with its practically evidenced intensity, only 1 520 artefacts could be existed in its cubic meter of layer). By means of an analogous experiment it has been identified that in case of the sites, where along with metal and glass artefacts, mainly diverse types of pottery could be unearthed, i.e. actually in case of the sites dated from the period following the Neolithic Age, theoretically maximal amount of artefacts in a cubic meter of layer could reach 24 000 artefacts. Thus, the difference in amounts of artefacts found at the sites dated from the Stone Age and the subsequent periods is apparent. Therefore, taking into account the previously mentioned data, from my point of view, a criterion with two different quantitative amounts should be elaborated on the basis of one and the same methodology both for the sites dated from the Stone Age (on the one hand), and for the ones dated from the following epochs (on the other hand).

Thus, we are aware that theoretically 172 000 units can exist in a cubic meter of the cultural layer (in case of sites dated from the Stone Age) and 24 000 ones (in case of the following epochs). Correspondingly, both poles of the criterion are defined: from 0 up to 172 000 (for the Stone Age) and from 0 up to 24 000 (for the following epochs), i.e. both aforementioned digits (172 000 and 24 000) represent invariable theoretical values, the constants.

Afterwards, the definition of the algorithm, by means of which the amount of the archaeological material in the layer - i.e. the intensity of the layer can be actually estimated - becomes simplified. The point is that spreading of archaeological material in a layer represents a certain probability since it implies a concrete amount of the possibilities out of all possibilities, i.e. occurrence of a concrete amount of the material in the layer out of all theoretically possible amounts. Therefore, for elaboration of a concrete algorithm we used the classical definition of the probability theory which expresses possibility of a concrete amount of possibilities out of all feasible possibilities. Thus, if we denote the constant, i.e. all possible amounts defined by us, by *m*, and the concrete possibility, i.e. a concrete amount of the material in the excavated habitation layer, by *n*, then the intensity of the layer could be calculated by means of the following simple algorithm:

$$P = n/m \quad (1)$$

But, since the criterion has to be bounded from both sides (from its minimum up to its maximum), as well as for the reason that in case of abundance of the amount of the material in the layer, interpretation of the numeric amount reaching 172 000 or even 24 000 is complicated, we have introduced a percentage amount of this algorithm, i.e.

$$P = 100 \, n/m \quad (2)$$

If we insert digital values of the existing constants, then the intensity of the layer in case of the sites dated from the Stone Age will correspond to:

$$P = n/1720 \, \%$$

If we round down the data, it will correspond to:

$$P = n/1700 \% \quad (3)$$

While for the subsequent epochs, it will correspond to:

$$P = n/240 \% \quad (4)$$

Thus, we have got the simplest formula estimating the intensity of the layer. Besides, it envisages other quantitative components of a layer – its area and its thickness. The numerator of this formula, i.e. value of *n* for a concrete layer actually should be calculated in the course of fieldwork (or later, on the basis of existing documentation), i.e. *n* denotes the amount of material discovered on each one square meter of area at the depth of one meter.

However, in some cases thickness of a layer is less than 1 meter. In case of the layer, the thickness of which is equal to 20 cm, a simplified type of the formula can be used:

$$P = n/340 \% \quad (5)$$

and

$$P = n/50 \% \quad (6)$$

Actually both formulae are extremely simple. But one more problem arises in the course of definition of the intensity of the layer. Namely, which of these two - a fragment or an intact item, for instance a vessel - should be considered as a single item. It should be also marked that mainly pottery is encountered in layers in the form of fragments. Therefore attention was focused on the study of the pottery. Each singularly excavated artifact, without regard for being a fragment or an intact item, should be considered as a single find. This conclusion is justified by the fact that, as it appears, occurrence (in other words, possibility) of breaking of any vessel into fragments at each settlement is approximately similar regardless of difference, existing between them due to their chronological assignments, sizes of pottery shards or degrees of their baking. However, if we separately consider vessels of one and the same type, it appears that they can break into fragments of various numbers or can stay intact. For instance, pots found at various sites were broken into 25, 50 or even 8 fragments. However, sometimes vessels (e.g. pots) of the same type, which were excavated at the same site, can be unearthed in intact forms but sometimes could be found broken into 18 or 39 fragments. In other words, in one and the same layer the vessels of one and the same type could be intact or broken into 18 and 50 fragments, or into 18 and 39 ones. Besides, the similar possibility of breaking is noted among vessels with different sizes and technological characteristics that were excavated at sites dated from various periods of time. For instance, the pithos of the Hellenistic period found at the Vani site, the height of which reached 120 cm, was broken into 96 fragments. But, the bronze period pot, found at Ulyianovka with the height reaching only 35 cm was broken into 140 fragments. Thus, it should be repeatedly underlined that relying on the aforementioned and similar data obtained by means of the analysis of the archaeological materials, it could be concluded that nearly similar possibility exists for breaking of the pottery shapes at sites dated from any period of time. Consequently, from our point of view, while using the criterion, any separate fragment or any intact artifact could be regarded as a single item, causing no practical divergence. However, I think this issue needs to be explored in more detail.

From the point of spreading of materials in a layer, one more detail should be taken into consideration as well. In the course of researches of the layers of the settlement it is possible to evidence a greater concentration of artifacts in one part of the layer compared to the other one.

In addition, due to various reasons in one part could be considerably greater amounts of fragmented materials compared with the other one. This fact can cause drawing of wrong conclusions concerning a total amount of materials in a layer prior a complete excavation of a site. To avoid this error, from my point of view, we should use the temporary or local size of the intensity of the layer, expressing an amount of the material existed in the partially excavated site and the complete intensity, i.e. the absolute one, expressing the intensity of a completely excavated site (layer). It goes without saying that the absolute intensity is equal of the sum of the arithmetic means of the local intensities.

Thus, the definition of the intensity of the layer in the course of archaeological fieldwork could be performed by means of several equivalent simple formulae (3-4 or 5-6) which deal with the layers of different intensities. Expression of materials obtained in layers by means of the presented percentage form creates an idea, considerably clearer and easier for perception, about the intensity of the layer. Correspondingly, it helps to gain an insight into former lifestyles via human activities performed in manufacture and household as well as in trading, ritual and cultural spheres.

It should be also marked that notwithstanding the fact that the definition of a quantitative amount of the materials, existing in the layer, represents merely a narrow, concrete issue of the layer theory, it is closely related to other aspects of the theory in question. Of them should be noted: origin of cultural layers, conformity between the thickness of a cultural layer and its intensity, interpretation of one and the same intensities of the sites dated from various periods, and other issues, investigation and solving of which are connected to multiple and long-term experiments.

I stand squarely behind the necessity of carrying out investigations of similar problems, since only this way helps to find natural regularities doubtlessly existing in archaeology. Their formulation in the shape of separate, stable laws seems both inevitable and, what is main, possible, similarly to those in the natural sciences.